

Namespace OsmToolkit

Classes

[Member](#)

A [Member](#) within a [Relation](#) instance.

[Node](#)

Represents a point of interest on earth.

[OsmCoordinateBounds](#)

Represents the latitude and longitude bounds for OSM-data.

[OsmData](#)

Contains lists including all [Node](#), [Way](#) and [Relation](#) instances registered from OSM-sources.

[OsmEntity](#)

Represents a map element.

[OsmHeader](#)

Represents the header of the OSM-data, describing where it was generated and information about copyright.

[OsmPath](#)

Represents a path of multiple [OsmEntity](#) instances between two [Node](#) instances.

[Relation](#)

Representation of a geographic feature.

[ServiceCollectionExtensions](#)

[User](#)

Contains user data about OSM-creators.

[Way](#)

Connection between two or more nodes.

Enums

[ReferenceType](#)

Contains [OsmEntity](#) reference types.

Class Member

Namespace: [OsmToolkit](#)

Assembly: OsmToolkit.dll

A [Member](#) within a [Relation](#) instance.

```
public class Member
```

Inheritance

[object](#)  ← Member

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) ,
[object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Constructors

Member(ReferenceType, long)

Initializes a new [Member](#) object without a specified role.

```
public Member(ReferenceType type, long referenceId)
```

Parameters

type [ReferenceType](#)

The specified [OsmEntity](#) type.

referenceId [long](#) 

The reference to the id of [OsmEntity](#), must be of values above 0.

Exceptions

[ArgumentOutOfRangeException](#) 

Throws if `referenceId` is less than or equal to 0.

Member(ReferenceType, long, string?)

Initializes a new [Member](#) object with a specified role.

```
public Member(ReferenceType type, long referenceId, string? role)
```

Parameters

type [ReferenceType](#)

The specified [OsmEntity](#) type.

referenceId [long](#)

The reference to the id of [OsmEntity](#), must be of values above 0.

role [string](#)

The specified [Member](#) role, can be null.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if `referenceId` is less than or equal to 0.

Properties

ReferenceId

A reference to the id of the [OsmEntity](#) instance.

```
[JsonPropertyName("ref")]  
[JsonPropertyOrder(1)]  
public long ReferenceId { get; }
```

Property Value

[long](#)

Role

The specified [Member](#) role for a [Relation](#).

```
[JsonPropertyName("role")]
[JsonPropertyOrder(2)]
public string Role { get; set; }
```

Property Value

[string](#)

Type

The specified [OsmEntity](#) type.

```
[JsonPropertyName("type")]
[JsonPropertyOrder(0)]
public ReferenceType Type { get; set; }
```

Property Value

[ReferenceType](#)

Class Node

Namespace: [OsmToolkit](#)

Assembly: OsmToolkit.dll

Represents a point of interest on earth.

```
public class Node : OsmEntity
```

Inheritance

[object](#)  ← [OsmEntity](#) ← Node

Inherited Members

[OsmEntity.Tags](#) , [OsmEntity.Id](#) , [OsmEntity.Visible](#) , [OsmEntity.Version](#) , [OsmEntity.ChangeSet](#) ,
[OsmEntity.Timestamp](#) , [OsmEntity.User](#) , [OsmEntity.AddTag\(string, string\)](#) , [OsmEntity.RemoveTag\(string\)](#) ,
[OsmEntity.HasTagKey\(string\)](#) , [OsmEntity.HasAnyTagKey\(IEnumerable<string>\)](#) ,
[OsmEntity.HasAllTagKeys\(IEnumerable<string>\)](#) , [OsmEntity.HasTagValue\(string\)](#) ,
[OsmEntity.HasAnyTagValue\(IEnumerable<string>\)](#) , [OsmEntity.HasAllTagValues\(IEnumerable<string>\)](#) ,
[OsmEntity.HasTag\(string, string\)](#) , [OsmEntity.HasAnyTag\(params \(string key, string value\)\[\]\)](#) ,
[OsmEntity.HasAnyTag\(Dictionary<string, string>\)](#) ,
[OsmEntity.HasAnyTag\(IEnumerable<KeyValuePair<string, string>>\)](#) ,
[OsmEntity.HasAllTags\(params \(string key, string value\)\[\]\)](#) ,
[OsmEntity.HasAllTags\(Dictionary<string, string>\)](#) ,
[OsmEntity.HasAllTags\(IEnumerable<KeyValuePair<string, string>>\)](#) , [object.Equals\(object\)](#)  ,
[object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  , [object.GetType\(\)](#)  ,
[object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  , [object.ToString\(\)](#) 

Constructors

Node(long, bool, int, long, DateTime, User, double, double)

Initializes a new [Node](#) object without specified tags.

```
public Node(long id, bool visible, int version, long changeSet, DateTime timestamp, User  
user, double latitude, double longitude)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

visible [bool](#)

Indicates whether the [OsmEntity](#) is currently visible or active. **true** if it is still part of the active map data, or **false** if otherwise.

version [int](#)

Current version number of [OsmEntity](#).

changeSet [long](#)

The id of the set of changes the [OsmEntity](#) was modified in.

timestamp [DateTime](#)

Last time [OsmEntity](#) was changed.

user [User](#)

Creator of [OsmEntity](#), can be **null**.

latitude [double](#)

Distance north or south of equator, must be within the range of -90 to 90.

longitude [double](#)

Distance east or west of prime meridian, must be within the range of -180 to 180.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if **id** is less than or equal to 0, or if **latitude** is out of the range -90 to 90 or **longitude** is out of the range -180 to 180.

Node(long, bool, int, long, DateTime, User, double, double, Dictionary<string, string>?)

Initializes a new [Node](#) object with specified tags.

```
[JsonConstructor]  
public Node(long id, bool visible, int version, long changeSet, DateTime timestamp, User  
user, double latitude, double longitude, Dictionary<string, string>? tags = null)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

visible [bool](#)

Indicates whether the [OsmEntity](#) is currently visible or active. **true** if it is still part of the active map data, or **false** if otherwise.

version [int](#)

Current version number of [OsmEntity](#).

changeSet [long](#)

The id of the set of changes the [OsmEntity](#) was modified in.

timestamp [DateTime](#)

Last time [OsmEntity](#) was changed.

user [User](#)

Creator of [OsmEntity](#), can be **null**.

latitude [double](#)

Distance north or south of equator, must be within the range of -90 to 90.

longitude [double](#)

Distance east or west of prime meridian, must be within the range of -180 to 180.

tags [Dictionary](#) <[string](#), [string](#)>

Optional dictionary containing key-value string pairs used to attach metadata to the [OsmEntity](#). If not provided or **null**, an empty dictionary is used by default.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if `id`, `version` or `changeSet` is less than or equal to 0, or if `latitude` is out of the range -90 to 90 or `longitude` is out of the range -180 to 180.

[ArgumentNullException](#)

Throws if `user` is `null`.

Node(long, double, double)

Initializes a new base [Node](#) without specified tags.

```
public Node(long id, double latitude, double longitude)
```

Parameters

`id` [long](#)

Unique identifier for [OsmEntity](#).

`latitude` [double](#)

Distance north or south of equator, must be within the range of -90 to 90.

`longitude` [double](#)

Distance east or west of prime meridian, must be within the range of -180 to 180.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if `id` is less than or equal to 0, or if `latitude` is out of the range -90 to 90 or `longitude` is out of the range -180 to 180.

Node(long, double, double, Dictionary<string, string>?)

Initializes a new base [Node](#) with specified tags.


```
public Node(long id, double latitude, double longitude, Dictionary<string, string>? tags
= null)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

latitude [double](#)

Distance north or south of equator, must be within the range of -90 to 90.

longitude [double](#)

Distance east or west of prime meridian, must be within the range of -180 to 180.

tags [Dictionary](#) <[string](#), [string](#)>

Optional dictionary containing key-value string pairs used to attach metadata to the [OsmEntity](#). If not provided or `null`, an empty dictionary is used by default.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if **id** is less than or equal to 0, or if **latitude** is out of the range -90 to 90 or **longitude** is out of the range -180 to 180.

Properties

Latitude

Distance north or south of equator, within the range of -90 to 90.

```
[JsonPropertyName("lat")]
[JsonPropertyOrder(6)]
public double Latitude { get; }
```

Property Value

[double](#)

Longitude

Distance east or west of prime meridian, within the range of -180 to 180.

```
[JsonPropertyName("lon")]  
[JsonPropertyOrder(7)]  
public double Longitude { get; }
```

Property Value

[double](#)

Methods

IsPartOf(Relation)

Checks if [Node](#) is part of a [Relation](#) instance.

```
public bool IsPartOf(Relation relation)
```

Parameters

relation [Relation](#)

[Relation](#) to check if [Node](#) is part of.

Returns

[bool](#)

true if [Node](#) is part of **relation**. Otherwise, **false**.

Exceptions

[ArgumentNullException](#)

Throws if **relation** is **null**.

IsPartOf(Way)

Checks if [Node](#) is part of a [Way](#) instance.

```
public bool IsPartOf(Way way)
```

Parameters

way [Way](#).

[Way](#) instance to check if [Node](#) is part of.

Returns

[bool](#)[↗]

true if [Node](#) is part of way. Otherwise, **false**.

Exceptions

[ArgumentNullException](#)[↗]

Throws if way is **null**.

IsPartOf(Way, Relation)

Checks if [Node](#) is part of both a [Way](#) instance and a [Relation](#) instance.

```
public bool IsPartOf(Way way, Relation relation)
```

Parameters

way [Way](#).

[Way](#) to check if [Node](#) is part of.

relation [Relation](#)

[Relation](#) to check if [Node](#) is part of.

Returns

[bool](#) 

`true` if [Node](#) is part of `way` and `relation`. Otherwise, `false`.

Exceptions

[ArgumentNullException](#) 

Throws if `way` or `relation` is `null`.

Class OsmCoordinateBounds

Namespace: [OsmToolkit](#)

Assembly: OsmToolkit.dll

Represents the latitude and longitude bounds for OSM-data.

```
public class OsmCoordinateBounds
```

Inheritance

[object](#)  ← OsmCoordinateBounds

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) ,
[object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Constructors

OsmCoordinateBounds(double, double, double, double)

Initializes a new instance of the [OsmCoordinateBounds](#) class.

```
public OsmCoordinateBounds(double minimumLatitude, double minimumLongitude, double  
maximumLatitude, double maximumLongitude)
```

Parameters

minimumLatitude [double](#) 

Minimum distance north or south of equator, within the range of -90 to 90.

minimumLongitude [double](#) 

Minimum distance east or west of prime meridian, within the range of -180 to 180.

maximumLatitude [double](#) 

Maximum distance north or south of equator, within the range of -90 to 90.

maximumLongitude [double](#) 

Maximum distance east or west of prime meridian, within the range of -180 to 180.

Exceptions

[ArgumentOutOfRangeException](#) 

Throws if `minimumLatitude` or `minimumLongitude` is out of the range -90 to 90, `minimumLongitude` or `maximumLongitude` is out of the range -180 to 180, `minimumLatitude` is equal to or greater than `maximumLatitude` or `minimumLongitude` is equal to or greater than `maximumLongitude`.

Properties

MaximumLatitude

Maximum distance north or south of equator, within the range of -90 to 90.

```
[JsonPropertyName("maxlat")]
[JsonPropertyOrder(2)]
public double MaximumLatitude { get; }
```

Property Value

[double](#) 

MaximumLongitude

Maximum distance east or west of prime meridian, within the range of -180 to 180.

```
[JsonPropertyName("maxlon")]
[JsonPropertyOrder(3)]
public double MaximumLongitude { get; }
```

Property Value

[double](#) 

MinimumLatitude

Minimum distance north or south of equator, within the range of -90 to 90.

```
[JsonPropertyName("minlat")]  
[JsonPropertyOrder(0)]  
public double MinimumLatitude { get; }
```

Property Value

[double](#)

MinimumLongitude

Minimum distance east or west of prime meridian, within the range of -180 to 180.

```
[JsonPropertyName("minlon")]  
[JsonPropertyOrder(1)]  
public double MinimumLongitude { get; }
```

Property Value

[double](#)

Class OsmData

Namespace: [OsmToolkit](#)

Assembly: OsmToolkit.dll

Contains lists including all [Node](#), [Way](#) and [Relation](#) instances registered from OSM-sources.

```
public class OsmData
```

Inheritance

[object](#)  ← OsmData

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) ,
[object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Constructors

OsmData(OsmHeader)

Initializes a new [OsmData](#) object without any registered instances of [OsmEntity](#).

```
public OsmData(OsmHeader header)
```

Parameters

header [OsmHeader](#)

The [OsmHeader](#) describing the header details for the OSM-data.

Exceptions

[ArgumentNullException](#) 

Throws if **header** is **null**.

OsmData(OsmHeader, OsmCoordinateBounds?, IEnumerable<Node>?, IEnumerable<Way>?, IEnumerable<Relation>?)

Initializes a new [OsmData](#) object with registered instances of class types [Node](#), [Way](#) and [Relation](#).

```
public OsmData(OsmHeader header, OsmCoordinateBounds? bounds = null, IEnumerable<Node>?  
nodes = null, IEnumerable<Way>? ways = null, IEnumerable<Relation>? relations = null)
```

Parameters

header [OsmHeader](#)

The [OsmHeader](#) describing the header details for the OSM-data.

bounds [OsmCoordinateBounds](#)

The [OsmCoordinateBounds](#) set for the OSM-data, can be `null`.

nodes [IEnumerable](#) [Node](#)

List of [Node](#) instances, can be `null`.

ways [IEnumerable](#) [Way](#)

List of [Way](#) instances, can be `null`.

relations [IEnumerable](#) [Relation](#)

List of [Relation](#) instances, can be `null`.

Exceptions

[ArgumentNullException](#)

Throws if **header** is `null`.

Properties

Bounds

[OsmCoordinateBounds](#) set for the OSM-data.

```
[JsonPropertyName("bounds")]
[JsonPropertyOrder(1)]
public OsmCoordinateBounds? Bounds { get; }
```

Property Value

[OsmCoordinateBounds](#)

Header

[OsmHeader](#) describing header details for the OSM-data.

```
[JsonPropertyName("header")]
[JsonPropertyOrder(0)]
public OsmHeader Header { get; set; }
```

Property Value

[OsmHeader](#)

Nodes

List of registered [Node](#) instances.

```
[JsonPropertyName("nodes")]
[JsonPropertyOrder(2)]
public IList<Node> Nodes { get; }
```

Property Value

[IList](#)  [Node](#)

Relations

List of registered [Relation](#) instances.

```
[JsonPropertyName("relations")]
[JsonPropertyOrder(4)]
public IList<Relation> Relations { get; }
```

Property Value

[IList](#) [<Relation>](#)

Ways

List of registered [Way](#) instances.

```
[JsonPropertyName("ways")]
[JsonPropertyOrder(3)]
public IList<Way> Ways { get; }
```

Property Value

[IList](#) [<Way>](#)

Methods

SortOsmData()

Sorts all OSM entities by their unique identifiers (Id) to ensure consistent serialization order.

```
public void SortOsmData()
```

Class OsmEntity

Namespace: [OsmToolkit](#)

Assembly: OsmToolkit.dll

Represents a map element.

```
public abstract class OsmEntity
```

Inheritance

[object](#)  ← OsmEntity

Derived

[Node](#), [Relation](#), [Way](#)

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) ,
[object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Constructors

OsmEntity(long)

Initializes a new default [OsmEntity](#) object with just a specified id.

```
protected OsmEntity(long id)
```

Parameters

id [long](#) 

Unique identifier for [OsmEntity](#), must be of values above 0.

Exceptions

[ArgumentOutOfRangeException](#) 

Throws if **id** is less than or equal to 0.

OsmEntity(long, bool, int, long, DateTime)

Initializes a new [OsmEntity](#) object with no tags.

```
protected OsmEntity(long id, bool visible, int version, long changeSet, DateTime timestamp)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#), must be of values above 0.

visible [bool](#)

Indicates whether the [OsmEntity](#) is currently visible or active, **true** if it is still part of the active map data, or **false** if otherwise.

version [int](#)

Current version number of [OsmEntity](#).

changeSet [long](#)

The id of the set of changes the [OsmEntity](#) was modified in.

timestamp [DateTime](#)

Last time [OsmEntity](#) was changed.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if **id** is less than or equal to 0.

OsmEntity(long, bool, int, long, DateTime, User?, Dictionary<string, string>?)

Initializes a new [OsmEntity](#) object with tags.

```
protected OsmEntity(long id, bool visible, int version, long changeSet, DateTime timestamp, User? user = null, Dictionary<string, string>? tags = null)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#), must be of values above 0.

visible [bool](#)

Indicates whether the [OsmEntity](#) is currently visible or active. **true** if it is still part of the active map data, or **false** if otherwise.

version [int](#)

Current version number of [OsmEntity](#).

changeSet [long](#)

The id of the set of changes the [OsmEntity](#) was modified in.

timestamp [DateTime](#)

Last time [OsmEntity](#) was changed.

user [User](#)

Creator of [OsmEntity](#), can be **null**.

tags [Dictionary](#) <[string](#), [string](#)>

Optional dictionary containing key-value string pairs used to attach metadata to the [OsmEntity](#). If not provided or **null**, an empty dictionary is used by default.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if **id** is less than or equal to 0.

OsmEntity(long, Dictionary<string, string>?)

Intializes a new default [OsmEntity](#) object with just a specified id and tags.

```
protected OsmEntity(long id, Dictionary<string, string>? tags = null)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#), must be of values above 0.

tags [Dictionary](#) <[string](#), [string](#)>

Optional dictionary containing key-value string pairs used to attach metadata to the [OsmEntity](#). If not provided or `null`, an empty dictionary is used by default.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if **id** is less than or equal to 0.

Properties

ChangeSet

Id of the set of changes the [OsmEntity](#) was last modified in.

```
[JsonPropertyName("changeset")]
[JsonPropertyOrder(3)]
public long ChangeSet { get; set; }
```

Property Value

[long](#)

Id

Unique identifier for [OsmEntity](#).

```
[JsonPropertyName("id")]
[JsonPropertyOrder(0)]
public long Id { get; }
```

Property Value

[long](#)

Tags

A read-only dictionary of key-value string pairs representing metadata for [OsmEntity](#).

```
[JsonPropertyName("tags")]
[JsonPropertyOrder(8)]
public IReadOnlyDictionary<string, string> Tags { get; }
```

Property Value

[IReadOnlyDictionary](#) <[string](#), [string](#)>

Timestamp

Last time [OsmEntity](#) was changed.

```
[JsonPropertyName("timestamp")]
[JsonPropertyOrder(4)]
public DateTime Timestamp { get; set; }
```

Property Value

[DateTime](#)

User

Creator of [OsmEntity](#).

```
[JsonPropertyName("user")]
[JsonPropertyOrder(5)]
public User? User { get; set; }
```

Property Value

[User](#)

Version

Current version number of [OsmEntity](#).

```
[JsonPropertyName("version")]  
[JsonPropertyOrder(2)]  
public int Version { get; set; }
```

Property Value

[int](#)

Visible

Indicates whether the [OsmEntity](#) is currently visible or active, **true** if it is still part of the active map data, or **false** if otherwise.

```
[JsonPropertyName("visible")]  
[JsonPropertyOrder(1)]  
public bool Visible { get; set; }
```

Property Value

[bool](#)

Methods

AddTag(string, string)

Adds or updates a dictionary containing metadata for [OsmEntity](#).

```
public void AddTag(string key, string value)
```

Parameters

key [string](#)

The key value to access the corresponding metadata, cannot be `null` or an empty string.

value [string](#)

The value describing the corresponding metadata, if `null` it will be set as an empty string.

Exceptions

[ArgumentException](#)

Throws if `key` is `null` or an empty string.

HasAllTagKeys(IEnumerable<string>)

Checks if the [OsmEntity](#) has tags with all of the specified `keys`.

```
public bool HasAllTagKeys(IEnumerable<string> keys)
```

Parameters

`keys` [IEnumerable](#) <[string](#)>

Keys to look for.

Returns

[bool](#)

`true`, if tags with all of the `keys` were found. Otherwise; `false`.

HasAllTagValues(IEnumerable<string>)

Checks if the [OsmEntity](#) has tags with all of the specified `values`.

```
public bool HasAllTagValues(IEnumerable<string> values)
```

Parameters

values [IEnumerable](#) <[string](#)>

Values to look for.

Returns

[bool](#)

true, if tags with all of the **values** were found. Otherwise; **false**.

HasAllTags(Dictionary<string, string>)

Checks if the [OsmEntity](#) has tags matching all of the specified **tags**.

```
public bool HasAllTags(Dictionary<string, string> tags)
```

Parameters

tags [Dictionary](#) <[string](#), [string](#)>

Tags to look for.

Returns

[bool](#)

true, if tags matching all of the **tags** were found. Otherwise; **false**.

HasAllTags(IEnumerable<KeyValuePair<string, string>>)

Checks if the [OsmEntity](#) has tags matching all of the specified **tags**.

```
public bool HasAllTags(IEnumerable<KeyValuePair<string, string>> tags)
```

Parameters

tags [IEnumerable](#) <[KeyValuePair](#) <[string](#), [string](#)>>

Tags to look for.

Returns

[bool](#)

`true`, if tags matching all of the `tags` were found. Otherwise; `false`.

HasAllTags(params (string key, string value)[])

Checks if the [OsmEntity](#) has tags matching all of the specified `tags`.

```
public bool HasAllTags(params (string key, string value)[] tags)
```

Parameters

`tags` ([string](#) `key`, [string](#) `value`)[]

Tags to look for.

Returns

[bool](#)

`true`, if tags matching all of the `tags` were found. Otherwise; `false`.

HasAnyTag(Dictionary<string, string>)

Checks if the [OsmEntity](#) has any tag matching one the specified `tags`.

```
public bool HasAnyTag(Dictionary<string, string> tags)
```

Parameters

`tags` [Dictionary](#) `<string, string>`

Tags to look for.

Returns

[bool](#)

`true`, if a tag with matches one of the `tags` was found. Otherwise; `false`.

HasAnyTag(IEnumerable<KeyValuePair<string, string>>)

Checks if the [OsmEntity](#) has any tag matching one the specified `tags`.

```
public bool HasAnyTag(IEnumerable<KeyValuePair<string, string>> tags)
```

Parameters

`tags` [IEnumerable](#) <[KeyValuePair](#) <[string](#), [string](#)>>

Tags to look for.

Returns

[bool](#)

`true`, if a tag with matches one of the `tags` was found. Otherwise; `false`.

HasAnyTag(params (string key, string value)[])

Checks if the [OsmEntity](#) has any tag matching one the specified `tags`.

```
public bool HasAnyTag(params (string key, string value)[] tags)
```

Parameters

`tags` ([string](#) [key](#), [string](#) [value](#))[]

Tags to look for.

Returns

[bool](#)

`true`, if a tag with matches one of the `tags` was found. Otherwise; `false`.

HasAnyTagKey(IEnumerable<string>)

Checks if the [OsmEntity](#) has a tag with any key of the specified **keys**.

```
public bool HasAnyTagKey(IEnumerable<string> keys)
```

Parameters

keys [IEnumerable](#) <[string](#)>

Keys to look for.

Returns

[bool](#)

true, if a tag with any of the **keys** was found. Otherwise; **false**.

HasAnyTagValue(IEnumerable<string>)

Checks if the [OsmEntity](#) has a tag with any value of the specified **values**.

```
public bool HasAnyTagValue(IEnumerable<string> values)
```

Parameters

values [IEnumerable](#) <[string](#)>

Values to look for.

Returns

[bool](#)

true, if a tag with any of the **values** was found. Otherwise; **false**.

HasTag(string, string)

Checks if the [OsmEntity](#) has a tag matching the specified key and value.

```
public bool HasTag(string key, string value)
```

Parameters

key [string](#)

Key to look for.

value [string](#)

Value to look for.

Returns

[bool](#)

true, if a tag with matches with **key** and **value** was found. Otherwise; **false**.

HasTagKey(string)

Checks if the [OsmEntity](#) has a tag with the specified **key**.

```
public bool HasTagKey(string key)
```

Parameters

key [string](#)

Key to look for.

Returns

[bool](#)

true, if a tag with **key** was found. Otherwise; **false**.

HasTagValue(string)

Checks if the [OsmEntity](#) has a tag with the specified **value**.

```
public bool HasTagValue(string value)
```

Parameters

value [string](#) 

Value to look for.

Returns

[bool](#) 

`true`, if a tag with `value` was found. Otherwise, `false`.

RemoveTag(string)

Removes a single dictionary containing metadata for [OsmEntity](#).

```
public bool RemoveTag(string key)
```

Parameters

key [string](#) 

The key value to access the corresponding metadata, cannot be `null` or an empty string.

Returns

[bool](#) 

`true` if removal is successful; otherwise, `false`.

Class OsmHeader

Namespace: [OsmToolkit](#)

Assembly: OsmToolkit.dll

Represents the header of the OSM-data, describing where it was generated and information about copyright.

```
public class OsmHeader
```

Inheritance

[object](#)  ← OsmHeader

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  , [object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  , [object.ToString\(\)](#) 

Constructors

OsmHeader(double)

Initiliazes a new default [OsmHeader](#) with version.

```
public OsmHeader(double version)
```

Parameters

version [double](#) 

Current version number of the OSM-data. Cannot be less than or equal to 0

Exceptions

[ArgumentOutOfRangeException](#) 

Throws if **version** is less than or equal to 0.

OsmHeader(double, string?, string?, string?, string?)

Initiliazes a new [OsmHeader](#) object.

```
public OsmHeader(double version, string? generator, string? copyright, string? attributionUrl, string? licenseUrl)
```

Parameters

version [double](#)

Current version number of the OSM-data. Cannot be less than or equal to 0

generator [string](#)

Generator used to create the OSM-data. If `null`, it will be set to "".

copyright [string](#)

Copyright owners of the OSM-data. If `null`, it will be set to "".

attributionUrl [string](#)

URL to the copyright owners' copyright page. If `null`, it will be set to "".

licenseUrl [string](#)

URL to the copyright owners' license of use. If `null`, it will be set to "".

Exceptions

[ArgumentOutOfRangeException](#)

Throws if `version` is less than or equal to 0.

Properties

AttributionUrl

Contains an URL to the copyright owners' copyright page.

```
[JsonPropertyName("attribution")]
[JsonPropertyOrder(3)]
public string AttributionUrl { get; }
```

Property Value

[string](#)

Copyright

Describes the copyright owners of the OSM-data.

```
[JsonPropertyName("copyright")]
[JsonPropertyOrder(2)]
public string Copyright { get; }
```

Property Value

[string](#)

Generator

Describes the generator used to create the OSM-data.

```
[JsonPropertyName("generator")]
[JsonPropertyOrder(1)]
public string Generator { get; }
```

Property Value

[string](#)

LicenseUrl

Contains an URL to the copyright owners' license of use.

```
[JsonPropertyName("license")]  
[JsonPropertyOrder(4)]  
public string LicenseUrl { get; }
```

Property Value

[string](#)

Version

Current version number of the OSM schema.

```
[JsonPropertyName("version")]  
[JsonPropertyOrder(0)]  
public double Version { get; }
```

Property Value

[double](#)

Class OsmPath

Namespace: [OsmToolkit](#)

Assembly: OsmToolkit.dll

Represents a path of multiple [OsmEntity](#) instances between two [Node](#) instances.

```
public class OsmPath
```

Inheritance

[object](#)  ← OsmPath

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) ,
[object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Constructors

OsmPath(OsmData, double)

Initializes a new and empty [OsmPath](#) object with no description.

```
public OsmPath(OsmData data, double totalDistance)
```

Parameters

data [OsmData](#)

Data containing [OsmEntity](#) instances making up the path.

totalDistance [double](#) 

Total distance of the path in meters.

OsmPath(OsmData, double, Node?, Node?)

Initializes a new [OsmPath](#) object with a startnode and endnode, but no description.

```
public OsmPath(OsmData data, double totalDistance, Node? startNode = null, Node? endNode = null)
```

Parameters

data [OsmData](#)

Data containing [OsmEntity](#) instances making up [OsmPath](#).

totalDistance [double](#) 

Total distance of [OsmPath](#) in meters.

startNode [Node](#)

Startpoint of [OsmPath](#), must already be in **data** and a [Way](#), otherwise it is set to **null**.

endNode [Node](#)

Endpoint of [OsmPath](#), must already be in **data** and a [Way](#), otherwise it is set to **null**

OsmPath(OsmData, double, Node?, Node?, string?)

Initializes a new [OsmPath](#) object with a description, startnode and endnode.

```
public OsmPath(OsmData data, double totalDistance, Node? startNode = null, Node? endNode = null, string? description = null)
```

Parameters

data [OsmData](#)

Data containing [OsmEntity](#) instances making up [OsmPath](#).

totalDistance [double](#) 

Total distance of [OsmPath](#) in meters.

startNode [Node](#)

Startpoint of [OsmPath](#), must already be in **data** and a [Way](#), otherwise it is set to **null**.

endNode [Node](#)

Endpoint of [OsmPath](#), must already be in `data` and a [Way](#), otherwise it is set to `null`.

description [string](#)↗

Optional description for additional information or messages.

OsmPath(OsmData, double, string?)

Initializes a new and empty [OsmPath](#) object with a description.

```
public OsmPath(OsmData data, double totalDistance, string? description = null)
```

Parameters

data [OsmData](#)

Data containing [OsmEntity](#) instances making up [OsmPath](#).

totalDistance [double](#)↗

Total distance of [OsmPath](#) in meters.

description [string](#)↗

Optional description for additional information or messages.

Properties

Data

Data containing all of the [OsmEntity](#) instances making up [OsmPath](#).

```
public OsmData Data { get; }
```

Property Value

[OsmData](#)

Description

Optional description containing additional information or messages when necessary.

```
public string? Description { get; }
```

Property Value

[string](#)

EndNode

[Node](#) instance representing the **endpoint** of [OsmPath](#), is **null** if path is empty.

```
public Node? EndNode { get; }
```

Property Value

[Node](#)

StartNode

[Node](#) instance representing the **startpoint** of [OsmPath](#), is **null** if path is empty.

```
public Node? StartNode { get; }
```

Property Value

[Node](#)

TotalDistance

Total distance of the path in meters, is 0 if path is empty.

```
public double TotalDistance { get; }
```


Property Value

[double](#) 

Enum ReferenceType

Namespace: [OsmToolkit](#)

Assembly: OsmToolkit.dll

Contains [OsmEntity](#) reference types.

```
public enum ReferenceType
```

Fields

```
node = 0
```

Represents members of class type [Node](#).

```
relation = 2
```

Represents members of class type [Relation](#).

```
way = 1
```

Represents members of class type [Way](#).

Class Relation

Namespace: [OsmToolkit](#)

Assembly: OsmToolkit.dll

Representation of a geographic feature.

```
public class Relation : OsmEntity
```

Inheritance

[object](#)  ← [OsmEntity](#) ← Relation

Inherited Members

[OsmEntity.Tags](#) , [OsmEntity.Id](#) , [OsmEntity.Visible](#) , [OsmEntity.Version](#) , [OsmEntity.ChangeSet](#) ,
[OsmEntity.Timestamp](#) , [OsmEntity.User](#) , [OsmEntity.AddTag\(string, string\)](#) , [OsmEntity.RemoveTag\(string\)](#) ,
[OsmEntity.HasTagKey\(string\)](#) , [OsmEntity.HasAnyTagKey\(IEnumerable<string>\)](#) ,
[OsmEntity.HasAllTagKeys\(IEnumerable<string>\)](#) , [OsmEntity.HasTagValue\(string\)](#) ,
[OsmEntity.HasAnyTagValue\(IEnumerable<string>\)](#) , [OsmEntity.HasAllTagValues\(IEnumerable<string>\)](#) ,
[OsmEntity.HasTag\(string, string\)](#) , [OsmEntity.HasAnyTag\(params \(string key, string value\)\[\]\)](#) ,
[OsmEntity.HasAnyTag\(Dictionary<string, string>\)](#) ,
[OsmEntity.HasAnyTag\(IEnumerable<KeyValuePair<string, string>>\)](#) ,
[OsmEntity.HasAllTags\(params \(string key, string value\)\[\]\)](#) ,
[OsmEntity.HasAllTags\(Dictionary<string, string>\)](#) ,
[OsmEntity.HasAllTags\(IEnumerable<KeyValuePair<string, string>>\)](#) , [object.Equals\(object\)](#)  ,
[object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  , [object.GetType\(\)](#)  ,
[object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  , [object.ToString\(\)](#) 

Constructors

Relation(long, bool, int, long, DateTime, User, IList<Member>)

Initializes a new [Relation](#) object without specified tags.

```
public Relation(long id, bool visible, int version, long changeSet, DateTime timestamp, User  
user, IList<Member> members)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

visible [bool](#)

Indicates whether the [OsmEntity](#) is currently visible or active. **true** if it is still part of the active map data, or **false** if otherwise.

version [int](#)

Current version number of [OsmEntity](#).

changeSet [long](#)

The id of the set of changes the [OsmEntity](#) was modified in.

timestamp [DateTime](#)

Last time [OsmEntity](#) was changed.

user [User](#)

Creator of [OsmEntity](#), can be **null**.

members [IList](#) <[Member](#)>

Collection of the [Relation](#)'s members.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if **id** is less than or equal to 0, or if **members** contain less than one element.

[ArgumentNullException](#)

Throws if **members** is **null**.

Relation(long, bool, int, long, DateTime, User, IList<Member>, Dictionary<string, string>?)

Initializes a new [Relation](#) object with specified tags.

```
public Relation(long id, bool visible, int version, long changeSet, DateTime timestamp, User user, IList<Member> members, Dictionary<string, string>? tags = null)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

visible [bool](#)

Indicates whether the [OsmEntity](#) is currently visible or active. **true** if it is still part of the active map data, or **false** if otherwise.

version [int](#)

Current version number of [OsmEntity](#).

changeSet [long](#)

The id of the set of changes the [OsmEntity](#) was modified in.

timestamp [DateTime](#)

Last time [OsmEntity](#) was changed.

user [User](#)

Creator of [OsmEntity](#), can be **null**..

members [IList](#) <[Member](#)>

Collection of the [Relation](#)'s members.

tags [Dictionary](#) <[string](#), [string](#)>

Optional dictionary containing key-value string pairs used to attach metadata to the [OsmEntity](#). If not provided or **null**, an empty dictionary is used by default.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if **id** is less than or equal to 0, or if **members** contain less than one element.

[ArgumentNullException](#)

Throws if `members` is `null`.

Relation(long, IList<Member>)

Initializes a new default [Relation](#) object without specified tags.

```
public Relation(long id, IList<Member> members)
```

Parameters

`id` [long](#)

Unique identifier for [OsmEntity](#).

`members` [IList](#) <[Member](#)>

Collection of the [Relation](#)'s members.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if `id` is less than or equal to 0, or if `members` contain less than one element.

[ArgumentNullException](#)

Throws if `members` is `null`.

Relation(long, IList<Member>, Dictionary<string, string>?)

Initializes a new default [Relation](#) object with specified tags.

```
public Relation(long id, IList<Member> members, Dictionary<string, string>? tags = null)
```

Parameters

`id` [long](#)

Unique identifier for [OsmEntity](#).

members [IList](#) <[Member](#)>

Collection of the [Relation](#)'s members.

tags [Dictionary](#) <[string](#), [string](#)>

Optional dictionary containing key-value string pairs used to attach metadata to the [OsmEntity](#). If not provided or `null`, an empty dictionary is used by default.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if `id` is less than or equal to 0, or if `members` contain less than one element.

[ArgumentNullException](#)

Throws if `members` is `null`.

Properties

Members

List of the [Relation](#) instance's members.

```
[JsonPropertyName("members")]  
[JsonPropertyOrder(6)]  
public IList<Member> Members { get; }
```

Property Value

[IList](#) <[Member](#)>

Class ServiceCollectionExtensions

Namespace: [OsmToolkit](#)

Assembly: OsmToolkit.dll

```
public static class ServiceCollectionExtensions
```

Inheritance

[object](#)  ← ServiceCollectionExtensions

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  , [object.GetType\(\)](#)  ,
[object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  , [object.ToString\(\)](#) 

Methods

AddOsmToolkit(IServiceCollection)

```
public static IServiceCollection AddOsmToolkit(this IServiceCollection services)
```

Parameters

services [IServiceCollection](#) 

Returns

[IServiceCollection](#) 

Class User

Namespace: [OsmToolkit](#)

Assembly: OsmToolkit.dll

Contains user data about OSM-creators.

```
public class User
```

Inheritance

[object](#)  ← User

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) ,
[object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Constructors

User(long, string)

Initializes a new [User](#) object.

```
public User(long id, string name)
```

Parameters

id [long](#) 

Unique identifier for [User](#).

name [string](#) 

Name of the [User](#).

Exceptions

[ArgumentOutOfRangeException](#) 

Throws if **id** is less than or equal to 0.

[ArgumentException](#)

Throws if `name` is `null` or an empty string.

Properties

Id

Unique identifier for [User](#).

```
[JsonPropertyName("id")]  
[JsonPropertyOrder(0)]  
public long Id { get; }
```

Property Value

[long](#)

Name

Name of the [User](#).

```
[JsonPropertyName("name")]  
[JsonPropertyOrder(1)]  
public string Name { get; set; }
```

Property Value

[string](#)

Class Way

Namespace: [OsmToolkit](#)

Assembly: OsmToolkit.dll

Connection between two or more nodes.

```
public class Way : OsmEntity
```

Inheritance

[object](#)  ← [OsmEntity](#) ← Way

Inherited Members

[OsmEntity.Tags](#) , [OsmEntity.Id](#) , [OsmEntity.Visible](#) , [OsmEntity.Version](#) , [OsmEntity.ChangeSet](#) ,
[OsmEntity.Timestamp](#) , [OsmEntity.User](#) , [OsmEntity.AddTag\(string, string\)](#) , [OsmEntity.RemoveTag\(string\)](#) ,
[OsmEntity.HasTagKey\(string\)](#) , [OsmEntity.HasAnyTagKey\(IEnumerable<string>\)](#) ,
[OsmEntity.HasAllTagKeys\(IEnumerable<string>\)](#) , [OsmEntity.HasTagValue\(string\)](#) ,
[OsmEntity.HasAnyTagValue\(IEnumerable<string>\)](#) , [OsmEntity.HasAllTagValues\(IEnumerable<string>\)](#) ,
[OsmEntity.HasTag\(string, string\)](#) , [OsmEntity.HasAnyTag\(params \(string key, string value\)\[\]\)](#) ,
[OsmEntity.HasAnyTag\(Dictionary<string, string>\)](#) ,
[OsmEntity.HasAnyTag\(IEnumerable<KeyValuePair<string, string>>\)](#) ,
[OsmEntity.HasAllTags\(params \(string key, string value\)\[\]\)](#) ,
[OsmEntity.HasAllTags\(Dictionary<string, string>\)](#) ,
[OsmEntity.HasAllTags\(IEnumerable<KeyValuePair<string, string>>\)](#) , [object.Equals\(object\)](#)  ,
[object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  , [object.GetType\(\)](#)  ,
[object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  , [object.ToString\(\)](#) 

Constructors

Way(long, bool, int, long, DateTime, User, List<long>)

Initializes a new [Way](#) object without specified tags.

```
public Way(long id, bool visible, int version, long changeSet, DateTime timestamp, User  
user, List<long> nodeReferenceIds)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

visible [bool](#)

Indicates whether the [OsmEntity](#) is currently visible or active. **true** if it is still part of the active map data, or **false** if otherwise.

version [int](#)

Current version number of [OsmEntity](#).

changeSet [long](#)

The id of the set of changes the [OsmEntity](#) was modified in.

timestamp [DateTime](#)

Last time [OsmEntity](#) was changed.

user [User](#)

Creator of [OsmEntity](#), can be **null**.

nodeReferenceIds [List](#) <[long](#)>

List of reference ids containing the [Way](#)'s nodes.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if **id** is less than or equal to 0, or if **nodeReferenceIds** contains less than two elements.

[ArgumentNullException](#)

Throws if **nodeReferenceIds** is **null**.

Way(long, bool, int, long, DateTime, User, List<long>, Dictionary<string, string>?)

Initializes a new [Way](#) object with specified tags.

```
public Way(long id, bool visible, int version, long changeSet, DateTime timestamp, User user, List<long> nodeReferenceIds, Dictionary<string, string>? tags = null)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

visible [bool](#)

Indicates whether the [OsmEntity](#) is currently visible or active. **true** if it is still part of the active map data, or **false** if otherwise.

version [int](#)

Current version number of [OsmEntity](#).

changeSet [long](#)

The id of the set of changes the [OsmEntity](#) was modified in.

timestamp [DateTime](#)

Last time [OsmEntity](#) was changed.

user [User](#)

Creator of [OsmEntity](#), can be **null**.

nodeReferenceIds [List](#) <[long](#)>

List of reference ids containing the [Way](#)'s nodes.

tags [Dictionary](#) <[string](#), [string](#)>

Optional dictionary containing key-value string pairs used to attach metadata to the [OsmEntity](#). If not provided or **null**, an empty dictionary is used by default.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if **id** is less than or equal to 0, or if **nodeReferenceIds** contains less than two elements.

[ArgumentNullException](#)

Throws if `nodeReferenceIds` is `null`.

Way(long, List<long>)

Initializes a new default [Way](#) object without specified tags.

```
public Way(long id, List<long> nodeReferenceIds)
```

Parameters

`id` [long](#)

Unique identifier for [OsmEntity](#).

`nodeReferenceIds` [List](#) <[long](#)>

List of reference ids containing the [Way](#)'s nodes.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if `id` is less than or equal to 0, or if `nodeReferenceIds` contains less than two elements.

[ArgumentNullException](#)

Throws if `nodeReferenceIds` is `null`.

Way(long, List<long>, Dictionary<string, string>?)

Initializes a new default [Way](#) object with specified tags.

```
public Way(long id, List<long> nodeReferenceIds, Dictionary<string, string>? tags = null)
```

Parameters

`id` [long](#)

Unique identifier for [OsmEntity](#).

nodeReferenceIds [List](#) <[long](#)>

List of reference ids containing the [Way](#)'s nodes.

tags [Dictionary](#) <[string](#), [string](#)>

Optional dictionary containing key-value string pairs used to attach metadata to the [OsmEntity](#). If not provided or **null**, an empty dictionary is used by default.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if **id** is less than or equal to 0, or if **nodeReferenceIds** contains less than two elements.

[ArgumentNullException](#)

Throws if **nodeReferenceIds** is **null**.

Properties

NodeReferenceIds

A read-only list of reference ids containing the [Way](#)'s nodes.

```
[JsonPropertyName("nodeRefs")]  
[JsonPropertyOrder(6)]  
public IReadOnlyList<long> NodeReferenceIds { get; }
```

Property Value

[IReadOnlyList](#) <[long](#)>

Methods

IsOneWay()

Checks if the [Way](#) is a oneway or not.

```
public bool IsOneWay()
```

Returns

[bool](#)

`true`, if it is a oneway. Otherwise; `false`.

Namespace OsmToolkit.Factories

Classes

[OsmEntityFactory](#)

Factory class responsible for creating instances of [OsmEntity](#), such as [Node](#), [Way](#) and [Relation](#).

Interfaces

[IOsmEntityFactory](#)

Defines factory methods for creating instances of [OsmEntity](#), such as [Node](#), [Way](#) and [Relation](#).

Interface IOsmEntityFactory

Namespace: [OsmToolkit.Factories](#)

Assembly: OsmToolkit.dll

Defines factory methods for creating instances of [OsmEntity](#), such as [Node](#), [Way](#) and [Relation](#).

```
public interface IOsmEntityFactory
```

Methods

CreateNode(long, bool, int, long, DateTime, User, double, double)

Initializes a new [Node](#) object without specified tags.

```
Node CreateNode(long id, bool visible, int version, long changeSet, DateTime timestamp, User user, double latitude, double longitude)
```

Parameters

id [long](#) 

Unique identifier for [OsmEntity](#).

visible [bool](#) 

Indicates whether the [OsmEntity](#) is currently visible or active. **true** if it is still part of the active map data, or **false** if otherwise.

version [int](#) 

Current version number of [OsmEntity](#).

changeSet [long](#) 

The id of the set of changes the [OsmEntity](#) was modified in.

timestamp [DateTime](#) 

Last time [OsmEntity](#) was changed.

user [User](#)

Creator of [OsmEntity](#), can be `null`..

latitude [double](#)

Distance north or south of equator, must be within the range of -90 to 90.

longitude [double](#)

Distance east or west of prime meridian, must be within the range of -180 to 180.

Returns

[Node](#)

A new instance of [Node](#) without specified tags.

CreateNode(long, bool, int, long, DateTime, User, double, double, Dictionary<string, string>?)

Initializes a new [Node](#) object with specified tags.

```
Node CreateNode(long id, bool visible, int version, long changeSet, DateTime timestamp, User user, double latitude, double longitude, Dictionary<string, string>? tags = null)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

visible [bool](#)

Indicates whether the [OsmEntity](#) is currently visible or active. `true` if it is still part of the active map data, or `false` if otherwise.

version [int](#)

Current version number of [OsmEntity](#).

changeSet [long](#)

The id of the set of changes the [OsmEntity](#) was modified in.

timestamp [DateTime](#)

Last time [OsmEntity](#) was changed.

user [User](#)

Creator of [OsmEntity](#), can be `null`..

latitude [double](#)

Distance north or south of equator, must be within the range of -90 to 90.

longitude [double](#)

Distance east or west of prime meridian, must be within the range of -180 to 180.

tags [Dictionary](#) <[string](#), [string](#)>

Optional dictionary containing key-value string pairs used to attach metadata to the [OsmEntity](#). If not provided or `null`, an empty dictionary is used by default.

Returns

[Node](#)

A new instance of [Node](#) with specified tags.

CreateNode(long, double, double)

Initializes a new base [Node](#) without specified tags.

```
Node CreateNode(long id, double latitude, double longitude)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

latitude [double](#)

Distance north or south of equator, must be within the range of -90 to 90.

longitude [double](#)

Distance east or west of prime meridian, must be within the range of -180 to 180.

Returns

[Node](#)

A new default instance of [Node](#) without specified tags.

CreateNode(long, double, double, Dictionary<string, string>?)

Initializes a new base [Node](#) with specified tags.

```
Node CreateNode(long id, double latitude, double longitude, Dictionary<string, string>? tags  
= null)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

latitude [double](#)

Distance north or south of equator, must be within the range of -90 to 90.

longitude [double](#)

Distance east or west of prime meridian, must be within the range of -180 to 180.

tags [Dictionary](#) <[string](#), [string](#)>

Optional dictionary containing key-value string pairs used to attach metadata to the [OsmEntity](#). If not provided or `null`, an empty dictionary is used by default.

Returns

[Node](#)

A new default instance of [Node](#) with specified tags.

CreateRelation(long, bool, int, long, DateTime, User, List<Member>)

Initializes a new [Relation](#) object without specified tags.

```
Relation CreateRelation(long id, bool visible, int version, long changeSet, DateTime timestamp, User user, List<Member> members)
```

Parameters

id [long](#) 

Unique identifier for [OsmEntity](#).

visible [bool](#) 

Indicates whether the [OsmEntity](#) is currently visible or active. `true` if it is still part of the active map data, or `false` if otherwise.

version [int](#) 

Current version number of [OsmEntity](#).

changeSet [long](#) 

The id of the set of changes the [OsmEntity](#) was modified in.

timestamp [DateTime](#) 

Last time [OsmEntity](#) was changed.

user [User](#)

Creator of [OsmEntity](#), can be `null`.

members [List](#)  [Member](#)

Collection of the [Relation](#)'s members.

Returns

[Relation](#)

A new instance of [Relation](#) without specified tags.

CreateRelation(long, bool, int, long, DateTime, User, List<Member>, Dictionary<string, string>?)

Initializes a new [Relation](#) object with specified tags.

```
Relation CreateRelation(long id, bool visible, int version, long changeSet, DateTime timestamp, User user, List<Member> members, Dictionary<string, string>? tags = null)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

visible [bool](#)

Indicates whether the [OsmEntity](#) is currently visible or active. **true** if it is still part of the active map data, or **false** if otherwise.

version [int](#)

Current version number of [OsmEntity](#).

changeSet [long](#)

The id of the set of changes the [OsmEntity](#) was modified in.

timestamp [DateTime](#)

Last time [OsmEntity](#) was changed.

user [User](#)

Creator of [OsmEntity](#), can be **null**.

members [List](#) <[Member](#)>

Collection of the [Relation](#)'s members.

tags [Dictionary](#) <[string](#), [string](#)>

Optional dictionary containing key-value string pairs used to attach metadata to the [OsmEntity](#). If not provided or **null**, an empty dictionary is used by default.

Returns

[Relation](#)

A new instance of [Relation](#) with specified tags.

CreateRelation(long, IList<Member>)

Initializes a new default [Relation](#) object without specified tags.

```
Relation CreateRelation(long id, IList<Member> members)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

members [IList](#) <[Member](#)>

Collection of the [Relation](#)'s members.

Returns

[Relation](#)

A new default instance of [Relation](#) without specified tags.

CreateRelation(long, IList<Member>, Dictionary<string, string>?)

Initializes a new default [Relation](#) object with specified tags.

```
Relation CreateRelation(long id, IList<Member> members, Dictionary<string, string>? tags)
```


`= null)`

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

members [IList](#) <[Member](#)>

Collection of the [Relation](#)'s members.

tags [Dictionary](#) <[string](#), [string](#)>

Optional dictionary containing key-value string pairs used to attach metadata to the [OsmEntity](#). If not provided or `null`, an empty dictionary is used by default.

Returns

[Relation](#)

A new default instance of [Relation](#) with specified tags.

CreateWay(long, bool, int, long, DateTime, User, List<long>)

Initializes a new [Way](#) object without specified tags.

```
Way CreateWay(long id, bool visible, int version, long changeSet, DateTime timestamp, User user, List<long> nodeReferenceIds)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

visible [bool](#)

Indicates whether the [OsmEntity](#) is currently visible or active. `true` if it is still part of the active map data, or `false` if otherwise.

version [int](#)

Current version number of [OsmEntity](#).

changeSet [long](#)

The id of the set of changes the [OsmEntity](#) was modified in.

timestamp [DateTime](#)

Last time [OsmEntity](#) was changed.

user [User](#)

Creator of [OsmEntity](#), can be `null`.

nodeReferenceIds [List](#) <[long](#)>

List of reference ids containing the [Way](#)'s nodes.

Returns

[Way](#).

A new instance of [Way](#) without specified tags.

CreateWay(long, bool, int, long, DateTime, User, List<long>, Dictionary<string, string>?)

Initializes a new [Way](#) object with specified tags.

```
Way CreateWay(long id, bool visible, int version, long changeSet, DateTime timestamp, User user, List<long> nodeReferenceIds, Dictionary<string, string>? tags = null)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

visible [bool](#)

Indicates whether the [OsmEntity](#) is currently visible or active. `true` if it is still part of the active map data, or `false` if otherwise.

version [int](#)

Current version number of [OsmEntity](#).

changeSet [long](#)

The id of the set of changes the [OsmEntity](#) was modified in.

timestamp [DateTime](#)

Last time [OsmEntity](#) was changed.

user [User](#)

Creator of [OsmEntity](#), can be `null`.

nodeReferenceIds [List](#) <[long](#)>

List of reference ids containing the [Way](#)'s nodes.

tags [Dictionary](#) <[string](#), [string](#)>

Optional dictionary containing key-value string pairs used to attach metadata to the [OsmEntity](#). If not provided or `null`, an empty dictionary is used by default.

Returns

[Way](#).

A new instance of [Way](#) with specified tags.

CreateWay(long, List<long>)

Initializes a new default [Way](#) object without specified tags.

```
Way CreateWay(long id, List<long> nodeReferenceIds)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

nodeReferenceIds [List](#) <[long](#)>

List of reference ids containing the [Way](#)'s nodes.

Returns

[Way](#).

A new default instance of [Way](#) without specified tags.

CreateWay(long, List<long>, Dictionary<string, string>?)

Initializes a new default [Way](#) object with specified tags.

```
Way CreateWay(long id, List<long> nodeReferenceIds, Dictionary<string, string>? tags = null)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

nodeReferenceIds [List](#) <[long](#)>

List of reference ids containing the [Way](#)'s nodes.

tags [Dictionary](#) <[string](#), [string](#)>

Optional dictionary containing key-value string pairs used to attach metadata to the [OsmEntity](#). If not provided or `null`, an empty dictionary is used by default.

Returns

[Way](#).

A new default instance of [Way](#) with specified tags.

Class OsmEntityFactory

Namespace: [OsmToolkit.Factories](#)

Assembly: OsmToolkit.dll

Factory class responsible for creating instances of [OsmEntity](#), such as [Node](#), [Way](#) and [Relation](#).

```
public class OsmEntityFactory : IOsmEntityFactory
```

Inheritance

[object](#)  ← OsmEntityFactory

Implements

[IOsmEntityFactory](#).

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) ,
[object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Constructors

OsmEntityFactory()

Initializes a new instance of the [OsmEntityFactory](#) class.

```
public OsmEntityFactory()
```

Methods

CreateNode(long, bool, int, long, DateTime, User, double, double)

Initializes a new [Node](#) object without specified tags.

```
public Node CreateNode(long id, bool visible, int version, long changeSet, DateTime  
timestamp, User user, double latitude, double longitude)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

visible [bool](#)

Indicates whether the [OsmEntity](#) is currently visible or active. **true** if it is still part of the active map data, or **false** if otherwise.

version [int](#)

Current version number of [OsmEntity](#).

changeSet [long](#)

The id of the set of changes the [OsmEntity](#) was modified in.

timestamp [DateTime](#)

Last time [OsmEntity](#) was changed.

user [User](#)

Creator of [OsmEntity](#), can be **null**.

latitude [double](#)

Distance north or south of equator, must be within the range of -90 to 90.

longitude [double](#)

Distance east or west of prime meridian, must be within the range of -180 to 180.

Returns

[Node](#)

A new instance of [Node](#) without specified tags.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if `id` is less than or equal to 0, or if `latitude` is out of the range -90 to 90 or `longitude` is out of the range -180 to 180.

CreateNode(long, bool, int, long, DateTime, User, double, double, Dictionary<string, string>?)

Initializes a new [Node](#) object with specified tags.

```
public Node CreateNode(long id, bool visible, int version, long changeSet, DateTime timestamp, User user, double latitude, double longitude, Dictionary<string, string>? tags = null)
```

Parameters

`id` [long](#)

Unique identifier for [OsmEntity](#).

`visible` [bool](#)

Indicates whether the [OsmEntity](#) is currently visible or active. `true` if it is still part of the active map data, or `false` if otherwise.

`version` [int](#)

Current version number of [OsmEntity](#).

`changeSet` [long](#)

The id of the set of changes the [OsmEntity](#) was modified in.

`timestamp` [DateTime](#)

Last time [OsmEntity](#) was changed.

`user` [User](#)

Creator of [OsmEntity](#), can be `null`.

`latitude` [double](#)

Distance north or south of equator, must be within the range of -90 to 90.

longitude [double](#)

Distance east or west of prime meridian, must be within the range of -180 to 180.

tags [Dictionary](#) <[string](#), [string](#)>

Optional dictionary containing key-value string pairs used to attach metadata to the [OsmEntity](#). If not provided or `null`, an empty dictionary is used by default.

Returns

[Node](#)

A new instance of [Node](#) with specified tags.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if **id** is less than or equal to 0, or if **latitude** is out of the range -90 to 90 or **longitude** is out of the range -180 to 180.

CreateNode(long, double, double)

Initializes a new base [Node](#) without specified tags.

```
public Node CreateNode(long id, double latitude, double longitude)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

latitude [double](#)

Distance north or south of equator, must be within the range of -90 to 90.

longitude [double](#)

Distance east or west of prime meridian, must be within the range of -180 to 180.

Returns

[Node](#)

A new default instance of [Node](#) without specified tags.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if `id` is less than or equal to 0, or if `latitude` is out of the range -90 to 90 or `longitude` is out of the range -180 to 180.

CreateNode(long, double, double, Dictionary<string, string>?)

Initializes a new base [Node](#) with specified tags.

```
public Node CreateNode(long id, double latitude, double longitude, Dictionary<string, string>? tags = null)
```

Parameters

`id` [long](#)

Unique identifier for [OsmEntity](#).

`latitude` [double](#)

Distance north or south of equator, must be within the range of -90 to 90.

`longitude` [double](#)

Distance east or west of prime meridian, must be within the range of -180 to 180.

`tags` [Dictionary](#) <[string](#), [string](#)>

Optional dictionary containing key-value string pairs used to attach metadata to the [OsmEntity](#). If not provided or `null`, an empty dictionary is used by default.

Returns

[Node](#)

A new default instance of [Node](#) with specified tags.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if **id** is less than or equal to 0, or if **latitude** is out of the range -90 to 90 or **longitude** is out of the range -180 to 180.

CreateRelation(long, bool, int, long, DateTime, User, List<Member>)

Initializes a new [Relation](#) object without specified tags.

```
public Relation CreateRelation(long id, bool visible, int version, long changeSet, DateTime timestamp, User user, List<Member> members)
```

Parameters

id [long](#) 

Unique identifier for [OsmEntity](#).

visible [bool](#) 

Indicates whether the [OsmEntity](#) is currently visible or active. **true** if it is still part of the active map data, or **false** if otherwise.

version [int](#) 

Current version number of [OsmEntity](#).

changeSet [long](#) 

The id of the set of changes the [OsmEntity](#) was modified in.

timestamp [DateTime](#) 

Last time [OsmEntity](#) was changed.

user [User](#)

Creator of [OsmEntity](#), can be **null**.

members [List](#) <[Member](#)>

Collection of the [Relation](#)'s members.

Returns

[Relation](#)

A new instance of [Relation](#) without specified tags.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if **id** is less than or equal to 0, or if **members** contain less than one element.

[ArgumentNullException](#)

Throws if **members** is **null**.

CreateRelation(long, bool, int, long, DateTime, User, List<Member>, Dictionary<string, string>?)

Initializes a new [Relation](#) object with specified tags.

```
public Relation CreateRelation(long id, bool visible, int version, long changeSet, DateTime timestamp, User user, List<Member> members, Dictionary<string, string>? tags = null)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

visible [bool](#)

Indicates whether the [OsmEntity](#) is currently visible or active. **true** if it is still part of the active map data, or **false** if otherwise.

version [int](#)

Current version number of [OsmEntity](#).

changeSet [long](#)

The id of the set of changes the [OsmEntity](#) was modified in.

timestamp [DateTime](#)

Last time [OsmEntity](#) was changed.

user [User](#)

Creator of [OsmEntity](#), can be `null`.

members [List](#) <[Member](#)>

Collection of the [Relation](#)'s members.

tags [Dictionary](#) <[string](#), [string](#)>

Optional dictionary containing key-value string pairs used to attach metadata to the [OsmEntity](#). If not provided or `null`, an empty dictionary is used by default.

Returns

[Relation](#)

A new instance of [Relation](#) with specified tags.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if `id` is less than or equal to 0, or if `members` contain less than one element.

[ArgumentNullException](#)

Throws if `members` is `null`.

CreateRelation(long, IList<Member>)

Initializes a new default [Relation](#) object without specified tags.

```
public Relation CreateRelation(long id, IList<Member> members)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

members [IList](#) <[Member](#)>

Collection of the [Relation](#)'s members.

Returns

[Relation](#)

A new default instance of [Relation](#) without specified tags.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if **id** is less than or equal to 0, or if **members** contain less than one element.

[ArgumentNullException](#)

Throws if **members** is **null**.

CreateRelation(long, IList<Member>, Dictionary<string, string>?)

Initializes a new default [Relation](#) object with specified tags.

```
public Relation CreateRelation(long id, IList<Member> members, Dictionary<string, string>? tags = null)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

members [IList](#) <[Member](#)>

Collection of the [Relation](#)'s members.

tags [Dictionary](#) [<string, string>](#)

Optional dictionary containing key-value string pairs used to attach metadata to the [OsmEntity](#). If not provided or `null`, an empty dictionary is used by default.

Returns

[Relation](#)

A new default instance of [Relation](#) with specified tags.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if `id` is less than or equal to 0, or if `members` contain less than one element.

[ArgumentNullException](#)

Throws if `members` is `null`.

CreateWay(long, bool, int, long, DateTime, User, List<long>)

Initializes a new [Way](#) object without specified tags.

```
public Way CreateWay(long id, bool visible, int version, long changeSet, DateTime timestamp,
    User user, List<long> nodeReferenceIds)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

visible [bool](#)

Indicates whether the [OsmEntity](#) is currently visible or active. `true` if it is still part of the active map data, or `false` if otherwise.

version [int](#)

Current version number of [OsmEntity](#).

changeSet [long](#)

The id of the set of changes the [OsmEntity](#) was modified in.

timestamp [DateTime](#)

Last time [OsmEntity](#) was changed.

user [User](#)

Creator of [OsmEntity](#), can be **null**.

nodeReferenceIds [List](#) <[long](#)>

List of reference ids containing the [Way](#)'s nodes.

Returns

[Way](#).

A new instance of [Way](#) without specified tags.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if **id** is less than or equal to 0, or if **nodeReferenceIds** contains less than two elements.

[ArgumentNullException](#)

Throws if **nodeReferenceIds** is **null**.

CreateWay(long, bool, int, long, DateTime, User, List<long>, Dictionary<string, string>?)

Initializes a new [Way](#) object with specified tags.

```
public Way CreateWay(long id, bool visible, int version, long changeSet, DateTime timestamp,
    User user, List<long> nodeReferenceIds, Dictionary<string, string>? tags = null)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

visible [bool](#)

Indicates whether the [OsmEntity](#) is currently visible or active. **true** if it is still part of the active map data, or **false** if otherwise.

version [int](#)

Current version number of [OsmEntity](#).

changeSet [long](#)

The id of the set of changes the [OsmEntity](#) was modified in.

timestamp [DateTime](#)

Last time [OsmEntity](#) was changed.

user [User](#)

Creator of [OsmEntity](#), can be **null**.

nodeReferenceIds [List](#) <[long](#)>

List of reference ids containing the [Way](#)'s nodes.

tags [Dictionary](#) <[string](#), [string](#)>

Optional dictionary containing key-value string pairs used to attach metadata to the [OsmEntity](#). If not provided or **null**, an empty dictionary is used by default.

Returns

[Way](#)

A new instance of [Way](#) with specified tags.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if **id** is less than or equal to 0, or if **nodeReferenceIds** contains less than two elements.

[ArgumentNullException](#)

Throws if `nodeReferenceIds` is `null`.

CreateWay(long, List<long>)

Initializes a new default [Way](#) object without specified tags.

```
public Way CreateWay(long id, List<long> nodeReferenceIds)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

nodeReferenceIds [List](#) <[long](#)>

List of reference ids containing the [Way](#)'s nodes.

Returns

[Way](#).

A new default instance of [Way](#) without specified tags.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if `id` is less than or equal to 0, or if `nodeReferenceIds` contains less than two elements.

[ArgumentNullException](#)

Throws if `nodeReferenceIds` is `null`.

CreateWay(long, List<long>, Dictionary<string, string>?)

Initializes a new default [Way](#) object with specified tags.

```
public Way CreateWay(long id, List<long> nodeReferenceIds, Dictionary<string, string>? tags  
= null)
```

Parameters

id [long](#)

Unique identifier for [OsmEntity](#).

nodeReferenceIds [List](#) <[long](#)>

List of reference ids containing the [Way](#)'s nodes.

tags [Dictionary](#) <[string](#), [string](#)>

Optional dictionary containing key-value string pairs used to attach metadata to the [OsmEntity](#). If not provided or `null`, an empty dictionary is used by default.

Returns

[Way](#)

A new default instance of [Way](#) with specified tags.

Exceptions

[ArgumentOutOfRangeException](#)

Throws if **id** is less than or equal to 0, or if **nodeReferenceIds** contains less than two elements.

[ArgumentNullException](#)

Throws if **nodeReferenceIds** is `null`.

Namespace OsmToolkit.Finders

Classes

[PathOptions](#)

Represents options for path traversal through [Way](#) instances.

Interfaces

[INearestNodesFinder](#)

Defines finder methods for finding the nearest **possible** [Node](#) instances.

[IOsmFinderV2<T>](#)

Defines finder methods for finding specific **T** instances in an [OsmData](#) instance. This interface replaces [IOsmFinder<T>](#) and provides additional features.

[IOsmFinder<T>](#)

Defines finder methods for finding specific **T** instances in an [OsmData](#) instance.

[IOsmValueFinder<T>](#)

Defines finder methods for finding [OsmEntity](#) instances by **tag** data.

[IShortestPathFinder](#)

Defines finder methods for finding the **shortest** possible path.

[IWithinDistanceFinder<T>](#)

Defines finder methods for finding nearby [OsmEntity](#) instances within a specified **radius**.

Enums

[TravelMode](#)

Selects forms of transport or travel to know what type of **highway:*** to filter out using [PathOptions](#). Follows https://wiki.openstreetmap.org/wiki/OSM_tags_for_routing/Access_restrictions#Norway[↗].

Interface INearestNodesFinder

Namespace: [OsmToolkit.Finders](#)

Assembly: OsmToolkit.dll

Defines finder methods for finding the nearest *possible* [Node](#) instances.

```
public interface INearestNodesFinder
```

Methods

FindNearbyNodes(OsmData, Node, int)

Finds a limited amount of nearby [Node](#) instances from a specified [Node](#)'s coordinate. Allows nodes from the same [Way](#) and [Relation](#).

```
IReadOnlyList<Node> FindNearbyNodes(OsmData data, Node node, int limit)
```

Parameters

data [OsmData](#)

Data for [INearestNodesFinder](#) to use and search through.

node [Node](#)

Node for [INearestNodesFinder](#) to search from.

limit [int](#)[↗]

Max limit of nearby [Node](#) instances to find.

Returns

[IReadOnlyList](#)[↗] <[Node](#)>

An [IReadOnlyList<T>](#)[↗] instance with a limited amount of [Node](#) instances.

FindNearbyNodes(OsmData, Node, int, Dictionary<string, string>?, bool, bool)

Finds a limited amount of nearby [Node](#) instances from a specified [Node](#)'s coordinate with optional tags and rules for allowing nodes from the [Way](#) and [Relation](#).

```
ICollection<Node> FindNearbyNodes(OsmData data, Node node, int limit, Dictionary<string, string>? tags = null, bool allowSameWay = true, bool allowSameRelation = true)
```

Parameters

data [OsmData](#)

Data for [INearestNodesFinder](#) to use and search through.

node [Node](#)

Node for [INearestNodesFinder](#) to search from.

limit [int](#)

Max limit of nearby [Node](#) instances to find.

tags [Dictionary](#) <[string](#), [string](#)>

Dictionary containing key-value string pairs describing the metadata of the [OsmEntity](#) instances.

allowSameWay [bool](#)

Whether multiple [Node](#) instances within the same [Way](#) can be included in the results.

allowSameRelation [bool](#)

Whether multiple [Node](#) instances within the same [Relation](#) can be included in the results.

Returns

[ICollection](#) <[Node](#)>

An [ICollection<T>](#) instance with a limited amount of [Node](#) instances.

FindNearbyNodes(OsmData, double, double, int)

Finds a limited amount of nearby [Node](#) instances from a specified coordinate. Allows nodes from the same [Way](#) and [Relation](#).

```
ICollection<Node> FindNearbyNodes(OsmData data, double latitude, double longitude,
int limit)
```

Parameters

data [OsmData](#)

Data for [INearestNodesFinder](#) to use and search through.

latitude [double](#)

Latitude for [INearestNodesFinder](#) to search from.

longitude [double](#)

Longitude for [INearestNodesFinder](#) to search from.

limit [int](#)

Max limit of nearby [Node](#) instances to find.

Returns

[ICollection](#) <[Node](#)>

An [ICollection<T>](#) instance with a limited amount of [Node](#) instances.

FindNearbyNodes(OsmData, double, double, int, Dictionary<string, string>?, bool, bool)

Finds a limited amount of nearby [Node](#) instances from a specified coordinate with optional tags and rules for allowing nodes from the [Way](#) and [Relation](#).

```
ICollection<Node> FindNearbyNodes(OsmData data, double latitude, double longitude, int
limit, Dictionary<string, string>? tags = null, bool allowSameWay = true, bool
allowSameRelation = true)
```

Parameters

data [OsmData](#)

Data for [INearestNodesFinder](#) to use and search through.

latitude [double](#)

Latitude for [INearestNodesFinder](#) to search from.

longitude [double](#)

Longitude for [INearestNodesFinder](#) to search from.

limit [int](#)

Max limit of nearby [Node](#) instances to find.

tags [Dictionary](#) <[string](#), [string](#)>

Dictionary containing key-value string pairs describing the metadata of the [OsmEntity](#) instances.

allowSameWay [bool](#)

Whether multiple [Node](#) instances within the same [Way](#) can be included in the results.

allowSameRelation [bool](#)

Whether multiple [Node](#) instances within the same [Relation](#) can be included in the results.

Returns

[IReadOnlyList](#) <[Node](#)>

An [IReadOnlyList<T>](#) instance with a limited amount of [Node](#) instances.

FindNearestNode(OsmData, Node)

Finds the nearest possible [Node](#) instance from a specified [Node](#)'s coordinate.

```
Node? FindNearestNode(OsmData data, Node node)
```

Parameters

data [OsmData](#)

Data for [INearestNodesFinder](#) to use and search through.

node [Node](#)

Node for [INearestNodesFinder](#) to search from.

Returns

[Node](#)

Nearest possible [Node](#) instance, otherwise; `null`.

FindNearestNode(OsmData, Node, Dictionary<string, string>)

Finds the nearest possible [Node](#) instance from a specified [Node](#)'s coordinate with a tags filter.

```
Node? FindNearestNode(OsmData data, Node node, Dictionary<string, string> tags)
```

Parameters

data [OsmData](#)

Data for [INearestNodesFinder](#) to use and search through.

node [Node](#)

Node for [INearestNodesFinder](#) to search from.

tags [Dictionary](#) <[string](#), [string](#)>

Dictionary containing key-value string pairs describing the metadata of the [OsmEntity](#) instances.

Returns

[Node](#)

Nearest possible [Node](#) instance, otherwise; `null`.

FindNearestNode(OsmData, double, double)

Finds the nearest possible [Node](#) instance from a specified coordinate.

```
Node? FindNearestNode(OsmData data, double latitude, double longitude)
```

Parameters

data [OsmData](#)

Data for [INearestNodesFinder](#) to use and search through.

latitude [double](#)[↗]

Latitude for [INearestNodesFinder](#) to search from.

longitude [double](#)[↗]

Longitude for [INearestNodesFinder](#) to search from.

Returns

[Node](#)

Nearest possible [Node](#) instance, otherwise; `null`.

FindNearestNode(OsmData, double, double, Dictionary<string, string>)

Finds the nearest possible [Node](#) instance from a specified coordinate with a tags filter.

```
Node? FindNearestNode(OsmData data, double latitude, double longitude, Dictionary<string, string> tags)
```

Parameters

data [OsmData](#)

Data for [INearestNodesFinder](#) to use and search through.

latitude [double](#)[↗]

Latitude for [INearestNodesFinder](#) to search from.

longitude [double](#)

Longitude for [INearestNodesFinder](#) to search from.

tags [Dictionary](#) <[string](#), [string](#)>

Dictionary containing key-value string pairs describing the metadata of the [OsmEntity](#) instances.

Returns

[Node](#)

Nearest possible [Node](#) instance, otherwise; `null`.

Interface IOsmFinderV2<T>

Namespace: [OsmToolkit.Finders](#)

Assembly: OsmToolkit.dll

Defines finder methods for finding specific [T](#) instances in an [OsmData](#) instance. This interface replaces [IOsmFinder<T>](#) and provides additional features.

```
public interface IOsmFinderV2<T> : IOsmValueFinder<T>, INearestNodesFinder,
    IWithinDistanceFinder<T>, IShortestPathFinder where T : OsmEntity
```

Type Parameters

T

Type of [OsmEntity](#) instances to find.

Inherited Members

[IOsmValueFinder<T>.FindByOsmId\(OsmData, long\)](#), [IOsmValueFinder<T>.FindByTag\(OsmData, string\)](#),
[IOsmValueFinder<T>.FindByTag\(OsmData, string, string\)](#),
[IOsmValueFinder<T>.FindByTags\(OsmData, Dictionary<string, string>\)](#),
[INearestNodesFinder.FindNearestNode\(OsmData, Node\)](#),
[INearestNodesFinder.FindNearestNode\(OsmData, Node, Dictionary<string, string>\)](#),
[INearestNodesFinder.FindNearestNode\(OsmData, double, double\)](#),
[INearestNodesFinder.FindNearestNode\(OsmData, double, double, Dictionary<string, string>\)](#),
[INearestNodesFinder.FindNearbyNodes\(OsmData, Node, int\)](#),
[INearestNodesFinder.FindNearbyNodes\(OsmData, Node, int, Dictionary<string, string>, bool, bool\)](#),
[INearestNodesFinder.FindNearbyNodes\(OsmData, double, double, int\)](#),
[INearestNodesFinder.FindNearbyNodes\(OsmData, double, double, int, Dictionary<string, string>, bool, bool\)](#),
[IWithinDistanceFinder<T>.FindNearByRadius\(OsmData, Node, double\)](#),
[IWithinDistanceFinder<T>.FindNearByRadius\(OsmData, double, double, double\)](#),
[IWithinDistanceFinder<T>.FindNearByPathDistance\(OsmData, Node, double\)](#),
[IWithinDistanceFinder<T>.FindNearByPathDistance\(OsmData, double, double, double\)](#),
[IShortestPathFinder.FindShortestPath\(OsmData, Node, Node\)](#),
[IShortestPathFinder.FindShortestPath\(OsmData, Node, Node, PathOptions\)](#),
[IShortestPathFinder.FindShortestPath\(OsmData, double, double, double, double\)](#),
[IShortestPathFinder.FindShortestPath\(OsmData, double, double, double, double, PathOptions\)](#)

Interface IOsmFinder<T>

Namespace: [OsmToolkit.Finders](#)

Assembly: OsmToolkit.dll

Defines finder methods for finding specific **T** instances in an [OsmData](#) instance.

```
[Obsolete("Use IOsmFinderV2 instead.", false)]  
public interface IOsmFinder<T> where T : OsmEntity
```

Type Parameters

T

Type of [OsmEntity](#) instances to find.

Methods

FindByTag(OsmData, string)

Finds **T** instances that contain a specified tag by only key.

```
OsmData FindByTag(OsmData data, string key)
```

Parameters

data [OsmData](#)

Data for [IOsmFinder<T>](#) to use and search through.

key [string](#) 

The key value to access the corresponding metadata, cannot be **null** or an empty string.

Returns

[OsmData](#)

An [OsmData](#) instance with **T** instances that contain the specified tag.

FindByTag(OsmData, string, string?)

Finds **T** instances that contain a specified tag by key and value.

```
OsmData FindByTag(OsmData data, string key, string? value = null)
```

Parameters

data [OsmData](#)

Data for [IOsmFinder<T>](#) to use and search through.

key [string](#)

The key value to access the corresponding metadata, cannot be **null** or an empty string.

value [string](#)

The value describing the corresponding metadata, can be **null**.

Returns

[OsmData](#)

An [OsmData](#) instance with **T** instances that contain the specified tag and value.

FindNearByPathDistance(OsmData, double, double, double)

Finds **T** instances from a specified coordinate that are within a specified path distance.

```
OsmData FindNearByPathDistance(OsmData data, double lat, double lon, double distanceMeters)
```

Parameters

data [OsmData](#)

Data for [IOsmFinder<T>](#) to use and search through.

lat [double](#)

Latitude for [IOsmFinder<T>](#) to search from.

lon [double](#)

Longitude for [IOsmFinder<T>](#) to search from.

distanceMeters [double](#)

Distance in meters for [IOsmFinder<T>](#) to search within.

Returns

[OsmData](#)

An [OsmData](#) instance with **T** instances that are within path distance.

FindNearByRadius(OsmData, double, double, double)

Finds **T** instances from a specified coordinate that are within a specified radius.

```
OsmData FindNearByRadius(OsmData data, double lat, double lon, double radiusMeters)
```

Parameters

data [OsmData](#)

Data for [IOsmFinder<T>](#) to use and search through.

lat [double](#)

Latitude for [IOsmFinder<T>](#) to search from.

lon [double](#)

Longitude for [IOsmFinder<T>](#) to search from.

radiusMeters [double](#)

Radius in meters for [IOsmFinder<T>](#) to search within.

Returns

[OsmData](#)

An [OsmData](#) instance with **T** instances that are within radius.

Interface IOsmValueFinder<T>

Namespace: [OsmToolkit.Finders](#)

Assembly: OsmToolkit.dll

Defines finder methods for finding [OsmEntity](#) instances by **tag** data.

```
public interface IOsmValueFinder<T> where T : OsmEntity
```

Type Parameters

T

Type of [OsmEntity](#) instances to find.

Methods

FindByOsmId(OsmData, long)

Finds **T** instances that have a specified id.

```
T? FindByOsmId(OsmData data, long id)
```

Parameters

data [OsmData](#)

Data for [IOsmValueFinder<T>](#) to use and search through.

id [long](#) 

Unique identifier for **T**, must be of values above 0.

Returns

T

An [OsmEntity](#) instance that have the specified id if found, otherwise **null**.

FindByTag(OsmData, string)

Finds **T** instances that contain a specified tag by only key.

```
OsmData FindByTag(OsmData data, string key)
```

Parameters

data [OsmData](#)

Data for [IOsmValueFinder<T>](#) to use and search through.

key [string](#) 

The key value to access the corresponding metadata, cannot be **null** or an empty string.

Returns

[OsmData](#)

An [OsmData](#) instance with **T** instances that contain the specified tag.

FindByTag(OsmData, string, string?)

Finds **T** instances that contain a specified tag by key and value.

```
OsmData FindByTag(OsmData data, string key, string? value = null)
```

Parameters

data [OsmData](#)

Data for [IOsmValueFinder<T>](#) to use and search through.

key [string](#) 

The key value to access the corresponding metadata, cannot be **null** or an empty string.

value [string](#) 

The value describing the corresponding metadata, can be **null**.

Returns

[OsmData](#)

An [OsmData](#) instance with **T** instances that contain the specified tag and value.

FindByTags(OsmData, Dictionary<string, string>)

Finds **T** instances that contain the specified tags in a dictionary with key-value string pairs.

```
OsmData FindByTags(OsmData data, Dictionary<string, string> tags)
```

Parameters

data [OsmData](#)

Data for [IOsmValueFinder<T>](#) to use and search through.

tags [Dictionary](#) [<string, string>](#)

Dictionary containing key-value string pairs describing the metadata of the **T** instances.

Returns

[OsmData](#)

An [OsmData](#) instance with **T** instances that contain the specified tags and values.

Interface IShortestPathFinder

Namespace: [OsmToolkit.Finders](#)

Assembly: OsmToolkit.dll

Defines finder methods for finding the **shortest** possible path.

```
public interface IShortestPathFinder
```

Methods

FindShortestPath(OsmData, Node, Node)

Finds the shortest possible path made up of [OsmEntity](#) instances from a specified **start** [Node](#)'s coordinate to a specified **target** [Node](#)'s coordinate with **default** [PathOptions](#).

```
OsmPath FindShortestPath(OsmData data, Node startNode, Node targetNode)
```

Parameters

data [OsmData](#)

Data for [IShortestPathFinder](#) to use and search through.

startNode [Node](#)

Node for [IShortestPathFinder](#) to start searching from.

targetNode [Node](#)

Node for [IShortestPathFinder](#) to try to reach.

Returns

[OsmPath](#)

An [OsmPath](#) instance with an [OsmData](#) instance containing the [OsmEntity](#) instances that make up the shortest valid path, the total distance in meters and **startnode** and **endnode** of the path. If no valid path

was found, an empty [OsmPath](#) instance is returned with a description describing that it could not find a valid path.

FindShortestPath(OsmData, Node, Node, PathOptions)

Finds the shortest possible path made up of [OsmEntity](#) instances from a specified **start** [Node](#)'s coordinate to a specified **target** [Node](#)'s coordinate with **custom** [PathOptions](#).

```
OsmPath FindShortestPath(OsmData data, Node startNode, Node targetNode,  
    PathOptions pathOptions)
```

Parameters

data [OsmData](#)

Data for [IShortestPathFinder](#) to use and search through.

startNode [Node](#)

Node for [IShortestPathFinder](#) to start searching from.

targetNode [Node](#)

Node for [IShortestPathFinder](#) to try to reach.

pathOptions [PathOptions](#)

Options defining which [Way](#) instances [IShortestPathFinder](#) can search through.

Returns

[OsmPath](#)

An [OsmPath](#) instance with an [OsmData](#) instance containing the [OsmEntity](#) instances that make up the shortest valid path, the total distance in meters and **startnode** and **endnode** of the path. If no valid path was found, an empty [OsmPath](#) instance is returned with a description describing that it could not find a valid path.

FindShortestPath(OsmData, double, double, double, double)

Finds the shortest possible path made up of [OsmEntity](#) instances from a specified **start** coordinate to a specified **target** coordinate with **default** [PathOptions](#).

```
OsmPath FindShortestPath(OsmData data, double startLatitude, double startLongitude, double targetLatitude, double targetLongitude)
```

Parameters

data [OsmData](#)

Data for [IShortestPathFinder](#) to use and search through.

startLatitude [double](#) 

Latitude for [IShortestPathFinder](#) to start searching from.

startLongitude [double](#) 

Longitude for [IShortestPathFinder](#) to start searching from.

targetLatitude [double](#) 

Latitude for [IShortestPathFinder](#) to try to reach.

targetLongitude [double](#) 

Longitude for [IShortestPathFinder](#) to try to reach.

Returns

[OsmPath](#)

An [OsmPath](#) instance with an [OsmData](#) instance containing the [OsmEntity](#) instances that make up the shortest valid path, the total distance in meters and **startnode** and **endnode** of the path. If no valid path was found, an empty [OsmPath](#) instance is returned with a description describing that it could not find a valid path.

FindShortestPath(OsmData, double, double, double, double, PathOptions)

Finds the shortest possible path made up of [OsmEntity](#) instances from a specified **start** coordinate to a specified **target** coordinate with **custom** [PathOptions](#).

```
OsmPath FindShortestPath(OsmData data, double startLatitude, double startLongitude, double targetLatitude, double targetLongitude, PathOptions pathOptions)
```

Parameters

data [OsmData](#)

Data for [IShortestPathFinder](#) to use and search through.

startLatitude [double](#)[↗]

Latitude for [IShortestPathFinder](#) to start searching from.

startLongitude [double](#)[↗]

Longitude for [IShortestPathFinder](#) to start searching from.

targetLatitude [double](#)[↗]

Latitude for [IShortestPathFinder](#) to try to reach.

targetLongitude [double](#)[↗]

Longitude for [IShortestPathFinder](#) to try to reach.

pathOptions [PathOptions](#)

Options defining which [Way](#) instances that [OsmData](#) can search through.

Returns

[OsmPath](#)

An [OsmPath](#) instance with an [OsmData](#) instance containing the [OsmEntity](#) instances that make up the shortest valid path, the total distance in meters and **startnode** and **endnode** of the path. If no valid path was found, an empty [OsmPath](#) instance is returned with a description describing that it could not find a valid path.

Interface IWithinDistanceFinder<T>

Namespace: [OsmToolkit.Finders](#)

Assembly: OsmToolkit.dll

Defines finder methods for finding nearby [OsmEntity](#) instances within a specified **radius**.

```
public interface IWithinDistanceFinder<T> where T : OsmEntity
```

Type Parameters

T

Type of [OsmEntity](#) instances to find.

Methods

FindNearByPathDistance(OsmData, Node, double)

Finds **T** instances from a specified [Node](#)'s coordinate that are within a specified path distance.

```
OsmData FindNearByPathDistance(OsmData data, Node node, double distanceMeters)
```

Parameters

data [OsmData](#)

Data for [IWithinDistanceFinder<T>](#) to use and search through.

node [Node](#)

Node for [IWithinDistanceFinder<T>](#) to search from.

distanceMeters [double](#) 

Distance in meters for [IWithinDistanceFinder<T>](#) to search within.

Returns

[OsmData](#)

An [OsmData](#) instance with **T** instances that are within path distance.

FindNearByPathDistance(OsmData, double, double, double)

Finds **T** instances from a specified coordinate that are within a specified path distance.

```
OsmData FindNearByPathDistance(OsmData data, double lat, double lon, double distanceMeters)
```

Parameters

data [OsmData](#)

Data for [IWithinDistanceFinder<T>](#) to use and search through.

lat [double](#) 

Latitude for [IWithinDistanceFinder<T>](#) to search from.

lon [double](#) 

Longitude for [IWithinDistanceFinder<T>](#) to search from.

distanceMeters [double](#) 

Distance in meters for [IWithinDistanceFinder<T>](#) to search within.

Returns

[OsmData](#)

An [OsmData](#) instance with **T** instances that are within path distance.

FindNearByRadius(OsmData, Node, double)

Finds **T** instances from a specified [Node](#)'s that are within a specified radius.

```
OsmData FindNearByRadius(OsmData data, Node node, double radiusMeters)
```


Parameters

data [OsmData](#)

Data for [IWithinDistanceFinder<T>](#) to use and search through.

node [Node](#)

Node for [IWithinDistanceFinder<T>](#) to search from.

radiusMeters [double](#) 

Radius in meters for [IWithinDistanceFinder<T>](#) to search within.

Returns

[OsmData](#)

An [OsmData](#) instance with **T** instances that are within radius.

FindNearByRadius(OsmData, double, double, double)

Finds **T** instances from a specified coordinate that are within a specified radius.

```
OsmData FindNearByRadius(OsmData data, double lat, double lon, double radiusMeters)
```

Parameters

data [OsmData](#)

Data for [IWithinDistanceFinder<T>](#) to use and search through.

lat [double](#) 

Latitude for [IWithinDistanceFinder<T>](#) to search from.

lon [double](#) 

Longitude for [IWithinDistanceFinder<T>](#) to search from.

radiusMeters [double](#) 

Radius in meters for [IWithinDistanceFinder<T>](#) to search within.

Returns

[OsmData](#)

An [OsmData](#) instance with **T** instances that are within radius.

Class PathOptions

Namespace: [OsmToolkit.Finders](#)

Assembly: OsmToolkit.dll

Represents options for path traversal through [Way](#) instances.

```
public class PathOptions
```

Inheritance

[object](#)  ← PathOptions

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) ,
[object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Constructors

PathOptions()

Initializes a new [PathOptions](#) object with **default** options.

```
public PathOptions()
```

PathOptions(TravelMode, bool)

Initializes a new [PathOptions](#) object.

```
public PathOptions(TravelMode mode, bool avoidMotorway)
```

Parameters

mode [TravelMode](#)

Form of transport or travel to determine the paths that can be traversed.

avoidMotorway [bool](#) 

Whether to avoid motorways or not.

Properties

AvoidMotorway

Whether to avoid motorways or not, with **false** as default on initialize.

```
public bool AvoidMotorway { get; init; }
```

Property Value

[bool](#)

Mode

[TravelMode](#) to determine which paths can be traversed, with **Any** as default on initialize.

```
public TravelMode Mode { get; init; }
```

Property Value

[TravelMode](#)

Enum TravelMode

Namespace: [OsmToolkit.Finders](#)

Assembly: OsmToolkit.dll

Selects forms of transport or travel to know what type of **highway:*** to filter out using [PathOptions](#). Follows https://wiki.openstreetmap.org/wiki/OSM_tags_for_routing/Access_restrictions#Norway.

```
public enum TravelMode
```

Fields

Any = 0

The default travel mode with no specific filters. (does allow pathing in opposite of one-way)

Bicycle = 2

Travel by bicycle adds filters to not path along disallowed paths.

Foot = 1

Travel by foot adds filters to not path along disallowed paths. (does allow pathing in opposite of one-way)

Moped = 3

Travel by moped adds filters to not path along disallowed paths.

MotorCar = 4

Travel by motorcar adds filters to not path along disallowed paths.

Namespace OsmToolkit.Serialization

Interfaces

[IOsmDeserializer](#)

Defines methods for deserializing OSM data into [OsmData](#) objects.

[IOsmSerializer](#)

Defines methods for serializing [OsmData](#) objects into OSM data source.

Interface IOsmDeserializer

Namespace: [OsmToolkit.Serialization](#)

Assembly: OsmToolkit.dll

Defines methods for deserializing OSM data into [OsmData](#) objects.

```
public interface IOsmDeserializer
```

Methods

DeserializeAsync(string, CancellationToken)

Asynchronously deserializes OSM content from a string.

```
Task<OsmData> DeserializeAsync(string input, CancellationToken cancellationToken = default)
```

Parameters

input [string](#) 

A string containing valid OSM data source.

cancellationToken [CancellationToken](#) 

A cancellation token that can be used to cancel the operation.

Returns

[Task](#)  <[OsmData](#)>

An [OsmData](#) object representing the deserialized OSM data.

DeserializeFromFileAsync(string, CancellationToken)

Asynchronously deserializes OSM data from the specified file path.

```
Task<OsmData> DeserializeFromFileAsync(string path, CancellationToken cancellationToken  
= default)
```

Parameters

path [string](#)

The path to the file to deserialize.

cancellationToken [CancellationToken](#)

A cancellation token that can be used to cancel the operation.

Returns

[Task](#) <[OsmData](#)>

An [OsmData](#) object representing the deserialized OSM data.

Exceptions

[ArgumentException](#)

Thrown when the file extension is not 'osm' or 'xml'.

DeserializeFromStreamAsync(Stream, CancellationToken)

Asynchronously deserializes OSM data from a stream.

```
Task<OsmData> DeserializeFromStreamAsync(Stream stream, CancellationToken cancellationToken  
= default)
```

Parameters

stream [Stream](#)

A readable [Stream](#) containing OSM data.

cancellationToken [CancellationToken](#)

A cancellation token that can be used to cancel the operation.

Returns

[Task](#)  [OsmData](#)

An [OsmData](#) object representing the deserialized OSM data.

Interface IOsmSerializer

Namespace: [OsmToolkit.Serialization](#)

Assembly: OsmToolkit.dll

Defines methods for serializing [OsmData](#) objects into OSM data source.

```
public interface IOsmSerializer
```

Methods

SerializeAsync(OsmData, CancellationToken)

Asynchronously serializes the given [OsmData](#) object to a string.

```
Task<string> SerializeAsync(OsmData data, CancellationToken cancellationToken = default)
```

Parameters

data [OsmData](#)

The OSM data to serialize.

cancellationToken [CancellationToken](#)

A cancellation token that can be used to cancel the operation.

Returns

[Task](#) <[string](#) >

A string containing the serialized OSM data source.

SerializeToFileAsync(OsmData, string, CancellationToken)

Asynchronously serializes the given [OsmData](#) object and writes the data source to a file.

```
Task SerializeToFileAsync(OsmData data, string path, CancellationToken cancellationToken  
= default)
```

Parameters

data [OsmData](#)

The OSM data to serialize.

path [string](#) 

The path to the file to write to.

cancellationToken [CancellationToken](#) 

A cancellation token that can be used to cancel the operation.

Returns

[Task](#) 

SerializeToStreamAsync(OsmData, Stream, CancellationToken)

Asynchronously serializes the given [OsmData](#) object and writes the data source to the provided stream.

```
Task SerializeToStreamAsync(OsmData data, Stream stream, CancellationToken cancellationToken  
= default)
```

Parameters

data [OsmData](#)

The OSM data to serialize.

stream [Stream](#) 

The stream to write the data source output to.

cancellationToken [CancellationToken](#) 

A cancellation token that can be used to cancel the operation.

Returns

[Task](#) 

Namespace OsmToolkit.Serialization.Json

Interfaces

[IOsmJsonDeserializer](#)

Represents a deserializer that parses OSM Json data into [OsmData](#) objects.

[IOsmJsonSerializer](#)

Represents a serializer that parses [OsmData](#) objects into OSM Json.

Interface IOsmJsonDeserializer

Namespace: [OsmToolkit.Serialization.Json](#)

Assembly: OsmToolkit.dll

Represents a deserializer that parses OSM Json data into [OsmData](#) objects.

```
public interface IOsmJsonDeserializer : IOsmDeserializer
```

Inherited Members

[IOsmDeserializer.DeserializeAsync\(string, CancellationToken\)](#) ,
[IOsmDeserializer.DeserializeFromStreamAsync\(Stream, CancellationToken\)](#) ,
[IOsmDeserializer.DeserializeFromFileAsync\(string, CancellationToken\)](#)

Interface IOsmJsonSerializer

Namespace: [OsmToolkit.Serialization.Json](#)

Assembly: OsmToolkit.dll

Represents a serializer that parses [OsmData](#) objects into OSM Json.

```
public interface IOsmJsonSerializer : IOsmSerializer
```

Inherited Members

[IOsmSerializer.SerializeAsync\(OsmData, CancellationTokent\)](#) ,

[IOsmSerializer.SerializeToFileAsync\(OsmData, string, CancellationTokent\)](#) ,

[IOsmSerializer.SerializeToStreamAsync\(OsmData, Stream, CancellationTokent\)](#).

Namespace OsmToolkit.Serialization.Xml

Interfaces

[IOsmXmlDeserializer](#)

Represents a deserializer that parses OSM XML data into [OsmData](#) objects.

[IOsmXmlSerializer](#)

Represents a serializer that parses [OsmData](#) objects into OSM XML.

Interface IOsmXmlDeserializer

Namespace: [OsmToolkit.Serialization.Xml](#)

Assembly: OsmToolkit.dll

Represents a deserializer that parses OSM XML data into [OsmData](#) objects.

```
public interface IOsmXmlDeserializer : IOsmDeserializer
```

Inherited Members

[IOsmDeserializer.DeserializeAsync\(string, CancellationToken\)](#),
[IOsmDeserializer.DeserializeFromStreamAsync\(Stream, CancellationToken\)](#),
[IOsmDeserializer.DeserializeFromFileAsync\(string, CancellationToken\)](#)

Interface IOsmXmlSerializer

Namespace: [OsmToolkit.Serialization.Xml](#)

Assembly: OsmToolkit.dll

Represents a serializer that parses [OsmData](#) objects into OSM XML.

```
public interface IOsmXmlSerializer : IOsmSerializer
```

Inherited Members

[IOsmSerializer.SerializeAsync\(OsmData, CancellationTokent\)](#) ,

[IOsmSerializer.SerializeToFileAsync\(OsmData, string, CancellationTokent\)](#) ,

[IOsmSerializer.SerializeToStreamAsync\(OsmData, Stream, CancellationTokent\)](#).